

Enterprise Windows Active Directory Infrastructure Lab

Architecture Documentation (Version 1)

1. Purpose

This document describes the logical architecture of an enterprise Windows Server 2022 infrastructure lab built for learning Windows Server administration and enterprise networking.

2. Architecture Overview

The environment consists of one Windows Server 2022 virtual machine configured as the Domain Controller (DC01) and one Windows 11 client (CLIENT01). The server provides Active Directory Domain Services (AD DS), DNS, and DHCP. CLIENT01 joins the Active Directory domain and authenticates against DC01.

3. Architecture Components

Component	Role	Purpose
Windows Server 2022 (DC01)	Domain Controller	Hosts AD DS, DNS and DHCP
Active Directory Domain Services	Identity	Centralized authentication and authorization
DNS Server	Infrastructure	Resolves hostnames within the domain
DHCP Server	Infrastructure	Automatically assigns IPv4 configuration
Windows 11 Client (CLIENT01)	Endpoint	Domain-joined enterprise workstation
Oracle VirtualBox	Virtualization	Hosts the virtual machines

4. Logical Architecture

DC01 (Windows Server 2022)
■ ■ ■ Active Directory Domain Services (AD DS)
■ ■ ■ DNS Server
■ ■ ■ DHCP Server



CLIENT01 (Windows 11)
• Domain Joined
• Receives IP Address from DHCP
• Uses DNS for Name Resolution

5. Data Flow

- CLIENT01 boots.
- DHCP assigns IP configuration.
- DNS resolves the domain controller.
- User authenticates through Active Directory.
- Group Policy is applied after successful authentication.

6. Validation Performed

- Active Directory Users and Computers verified.
- DNS Manager verified.
- DHCP Address Leases verified.
- Windows 11 client successfully joined to the domain.
- Connectivity verified using **ping DC01**.
- Network configuration verified using **ipconfig /all**.
- Group Policy Management verified.

7. Skills Demonstrated

Windows Server Administration, Active Directory, DNS Administration, DHCP Administration, Enterprise Networking, Windows Client Administration, Infrastructure Deployment, Troubleshooting.

8. Future Enhancements

Future versions of this lab will integrate OPNsense Firewall, Wazuh SIEM, Splunk, Sysmon, Windows Event Forwarding (WEF), centralized log analysis, and threat detection capabilities.